

# Feature Selection Techniques

اختيار الميزات — Data Mining (CS602)

Candidate: Master of Computer Science Student

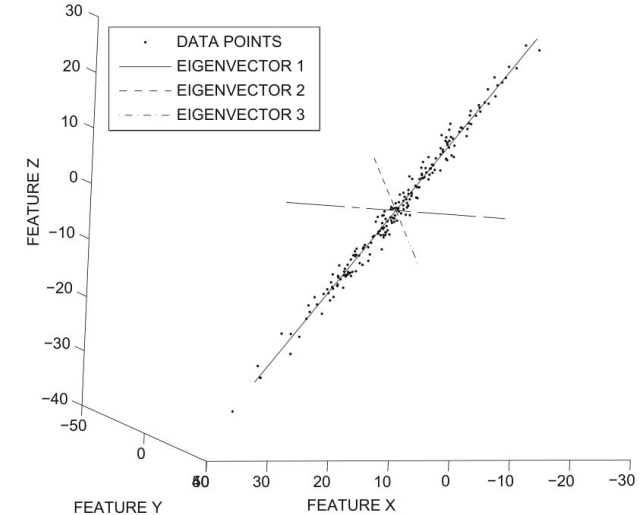
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Institution: College of Computer Science & IT — University of Wasit

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# Most columns in a real dataset do not earn their place.

- Irrelevant: no relationship to the target — a customer's telephone number.
- Redundant: duplicates information already present — age versus date of birth.
- Cost: poor-quality patterns, and a slower mining process.



*Highly correlated (redundant) data — the redundancy problem (Aggarwal, p.41)*

# Feature selection chooses existing columns; it never creates new ones.

- Select: keep a subset of the existing attributes (drop Phone).
- Construct: add a new column from existing ones ( $\text{area} = \text{height} \times \text{width}$ ).
- Extract: build a new representation with new axes (PCA components).
- All three sit under the umbrella of data reduction.

# The goal is the smallest subset that still preserves the class distribution.

*"The goal of attribute subset selection is to find a minimum set of attributes such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes." — Han, Kamber & Pei, p.104 "*

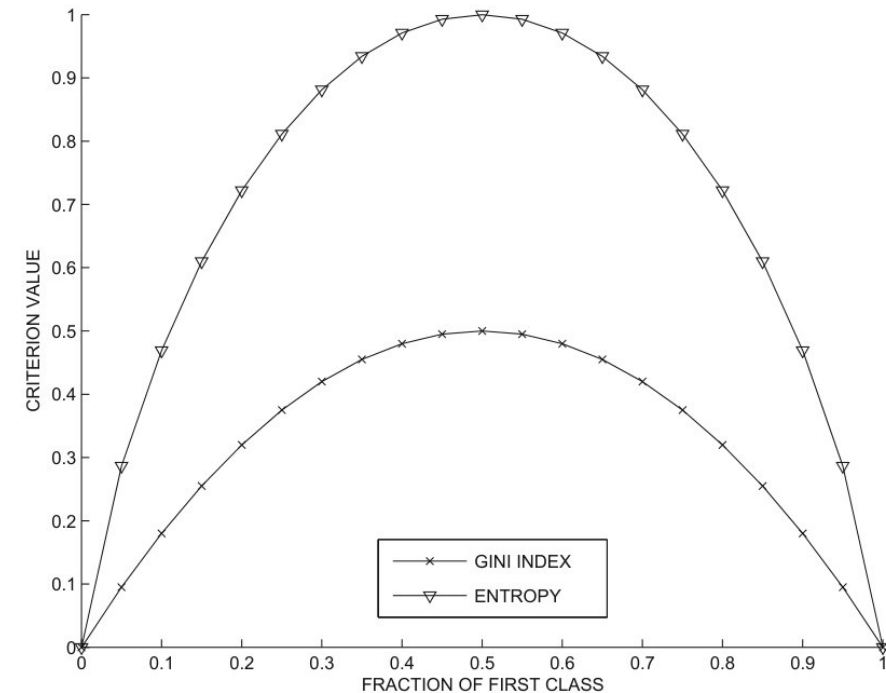
- Minimum set: we are minimising.
- Class distribution preserved: we are constrained.
- Together: a trade-off, not a one-sided goal.

# **There are 2-to-the-n possible subsets, so exhaustive search is impossible.**

- 20 features: 1,048,576 subsets. 30 features: over a billion. 50 features: over a quadrillion.
- H&K: an exhaustive search "can be prohibitively expensive".
- So we search heuristically — greedy: a locally optimal choice, hoping for a near-global optimum.

# Both Gini and entropy ask one question: how mixed are the classes?

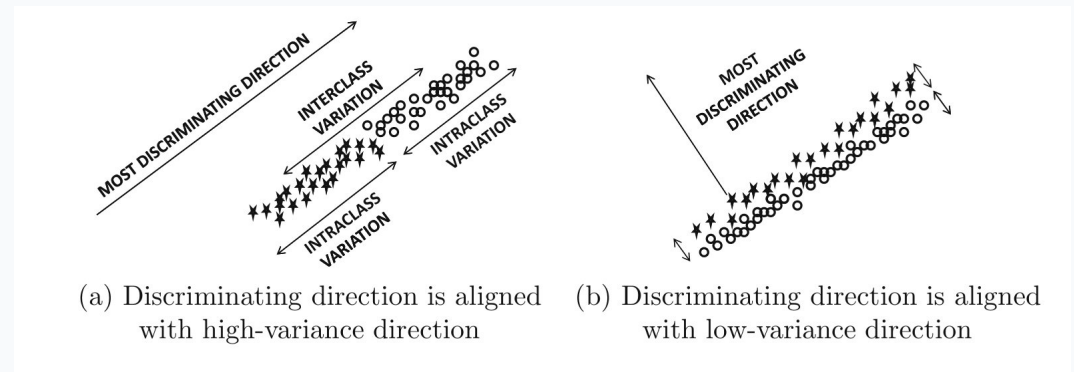
- Gini (Eq. 10.1): 1 minus the sum of squared class fractions.
- Entropy (Eq. 10.3): minus the sum of  $p \log_2 p$ .
- Both: 0 = perfect separation; higher = more mixing. Lower is better.



*Gini and entropy against class-distribution skew (Aggarwal, p.289)*

# The Fisher score rewards classes that are far apart and tight within.

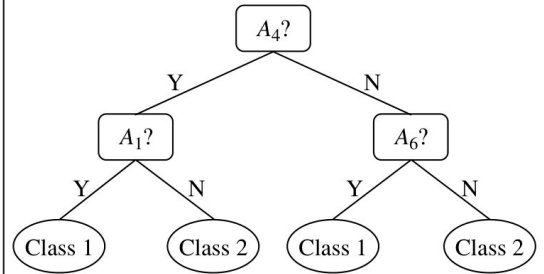
- Eq. 10.5: interclass separation divided by intraclass spread.
- Numerator: how far each class mean sits from the global mean.
- Denominator: how wide each class is. Higher is better.



*Fisher's discriminating direction vs class distribution (Aggarwal, p.291)*

# Four greedy searches all converge on the same reduced subset.

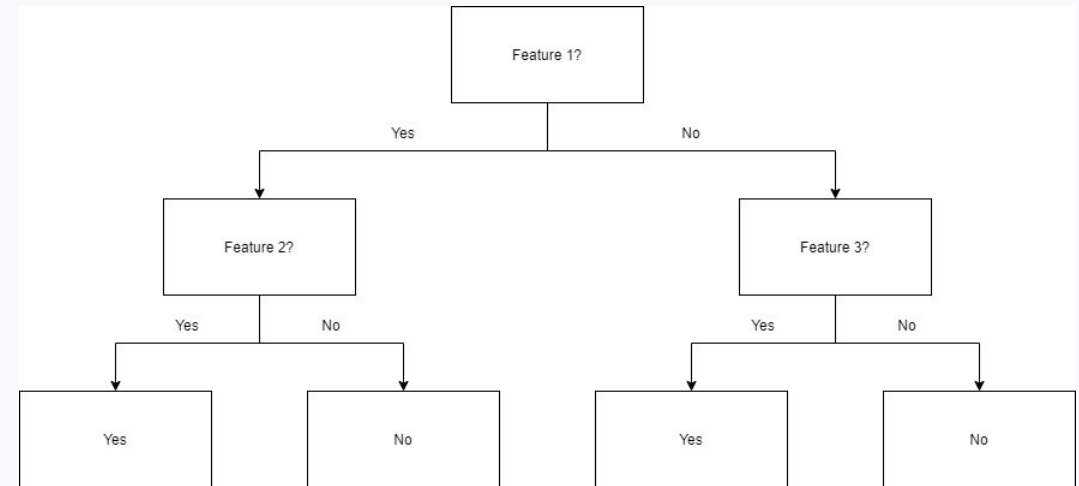
- Forward selection: start empty, add the best each step.
- Backward elimination: start full, remove the worst each step.
- Combination: add the best and remove the worst. Decision tree: keep the attributes it splits on.

| Forward selection                                                                                                                                                                                                                                                                                       | Backward elimination                                                                                                                                                                                                                                                                   | Decision tree induction                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Initial attribute set:<br/><math>\{A_1, A_2, A_3, A_4, A_5, A_6\}</math></p> <p>Initial reduced set:<br/><math>\{\}</math><br/><math>\Rightarrow \{A_1\}</math><br/><math>\Rightarrow \{A_1, A_4\}</math><br/><math>\Rightarrow</math> Reduced attribute set:<br/><math>\{A_1, A_4, A_6\}</math></p> | <p>Initial attribute set:<br/><math>\{A_1, A_2, A_3, A_4, A_5, A_6\}</math></p> <p><math>\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}</math><br/><math>\Rightarrow \{A_1, A_4, A_5, A_6\}</math><br/><math>\Rightarrow</math> Reduced attribute set:<br/><math>\{A_1, A_4, A_6\}</math></p> | <p>Initial attribute set:<br/><math>\{A_1, A_2, A_3, A_4, A_5, A_6\}</math></p>  <pre>graph TD; A4["A4?"] -- Y --&gt; A1["A1?"]; A4 -- N --&gt; A6["A6?"]; A1 -- Y --&gt; C1_1((Class 1)); A1 -- N --&gt; C2_1((Class 2)); A6 -- Y --&gt; C1_2((Class 1)); A6 -- N --&gt; C2_2((Class 2));</pre> <p><math>\Rightarrow</math> Reduced attribute set:<br/><math>\{A_1, A_4, A_6\}</math></p> |

*Greedy methods for attribute subset selection (Han, Kamber & Pei, p.104)*

# A decision tree keeps only the attributes it actually splits on.

- Each internal node is a test on an attribute.
- Each branch is an outcome; each leaf is a class prediction.
- Attributes that never appear in the tree are assumed irrelevant.

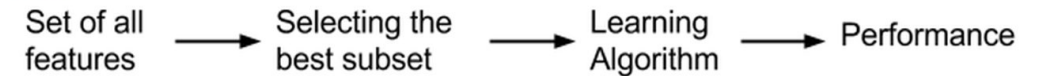


*A decision tree: internal tests, branches, and class leaves*

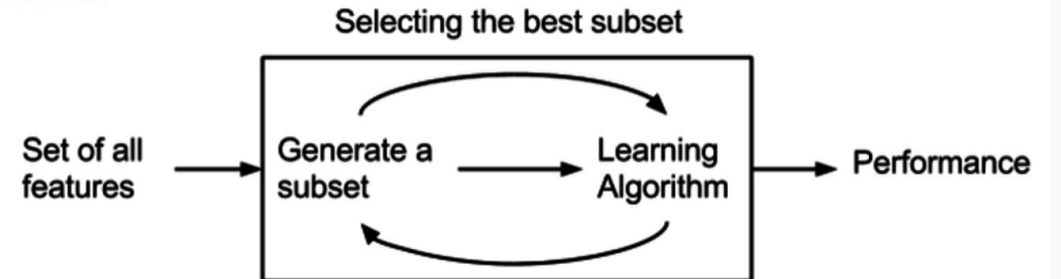
# Who computes the score? That single question defines three families.

- Filter: a fixed mathematical criterion — no classifier, cheap, model-agnostic.
- Wrapper: the classifier's own accuracy judges each subset — expensive, model-tied.
- Embedded: the model reveals the features while training — e.g. small weights, Lasso, RFE.

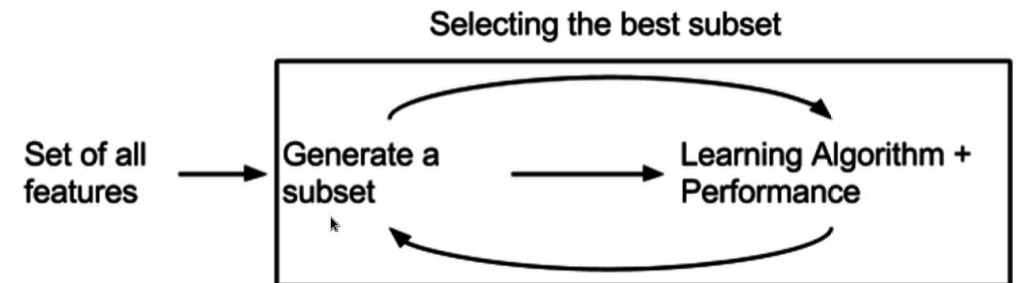
## FILTER



## WRAPPER



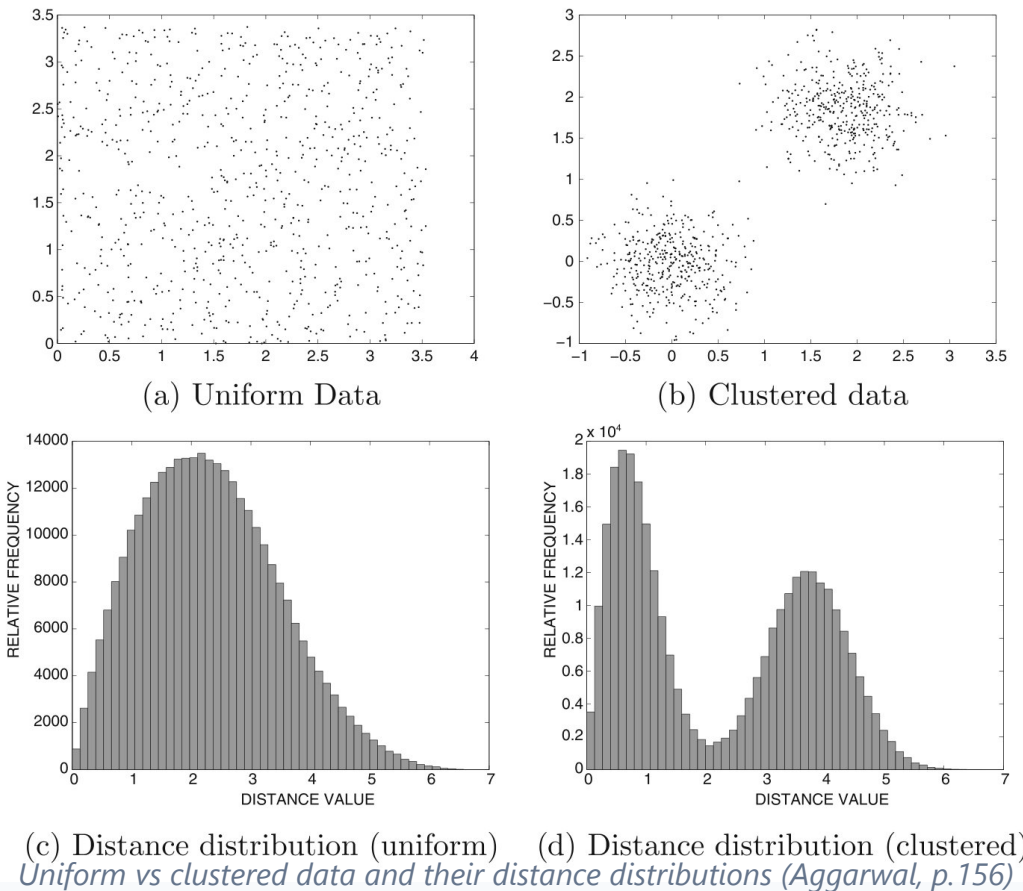
## EMBEDDED



*Filter, wrapper and embedded selection models*

# Without labels, the score must measure clustering tendency instead.

- Uniform data: one bell-shaped distance distribution. Clustered data: two peaks.
- Pick the subset that minimises the distance-distribution entropy.
- Named measures: term strength, classification-based relevance, Hopkins statistic.



# Sometimes the right feature does not exist yet — so build it.

*"In some cases, we may want to create new attributes based on others. Such attribute construction can help improve accuracy and understanding of structure in high-dimensional data." — Han, Kamber & Pei, p.105 "*

- Example:  $\text{area} = \text{height} \times \text{width}$  — a new column on the same axes.
- Selection removes; construction adds a column; extraction changes the axes.

## Five traps decide this topic in the exam.

- Selection chooses existing; extraction and construction create.
- Gini and entropy: lower is better. Fisher score: higher is better.
- Filter is model-free; wrapper is model-tied.
- Fisher score selects a feature; Fisher's discriminant builds a direction.
- Hopkins near 1 means clustered; 0.5 means uniform.

# The whole topic in five lines.

- Too many features hurt — noise and redundancy.
- We want the smallest subset that preserves class information.
- There are  $2^n$  subsets, so we score and search greedily.
- The score is a filter, or model-tied (wrapper or embedded).
- Selection chooses; construction and extraction create.

# References and image credits

- Han, Jiawei; Kamber, Micheline; Pei, Jian. Data Mining: Concepts and Techniques, 3rd ed. Morgan Kaufmann, 2011 — Section 3.4.4 (pp. 103-105).
- Aggarwal, Charu C. Data Mining: The Textbook. Springer, 2015 — Section 10.2 (pp. 287-293); Section 6.2 (pp. 155-158).
- Figures 3.6, 10.1, 10.2, 6.1, 2.2 reproduced from the two textbooks above, for study use, with page anchors.
- Decision-tree diagram: CollaborativeGeneticist, CC BY-SA 4.0, via Wikimedia Commons.
- Filter / wrapper / embedded diagrams: Lucien Mousin and Lastdreamer7591, CC BY-SA 4.0, via Wikimedia Commons.